

RFI 36C1026Q0218

Attachment 1

Telecommunications and IT Infrastructure

Standardized telecommunications and IT infrastructure components suitable for installation in VA data center and telecommunications spaces. All items shall conform to the Infrastructure Standard for Telecommunications Spaces (ISTS) and PG-18-12 Telecommunications Design Guide.

Product Category	Salient Characteristic	Requirement
Equipment Racks, Cabinets, and Enclosures	Form Factor	Standard 19-inch EIA-310-E compliant
	Height	42U minimum for data center; 24U–45U for telecom rooms per site conditions
	Load Capacity	Static load rating minimum 2,500 lbs
	Seismic	Seismic anchoring and bracing per VA seismic zone requirements
	Bundled Services	Site survey, delivery, installation, grounding, labeling, as-built documentation
Ladder Rack, Cable Tray, and Overhead Support	Type	Ladder rack, basket tray, or cable tray per site design
	Material	Aluminum or steel, appropriate for environment

Product Category	Salient Characteristic	Requirement
	Bundled Services	Pathway design, installation, firestopping, labeling, as-built documentation
Copper Structured Cabling Systems	Category	Minimum Category 6A per ISTS
	Standard	TIA-568 compliant; tested and certified end-to-end
	Bundled Services	Design, installation, termination, testing, certification, labeling, as-builts
Fiber Optic Structured Cabling Systems	Type	Single-mode and/or multimode per application requirements
	Connector	LC duplex or MPO/MTP as applicable
	Bundled Services	Design, installation, splicing/termination, OTDR testing, certification, as-builts
Patch Panels, Fiber Enclosures, and Termination Hardware	Compatibility	Compatible with installed cabling plant
	Bundled Services	Installation, labeling, cross-connect documentation

Product Category	Salient Characteristic	Requirement
Copper and Fiber Patch Cords	Category/Type	Matching or exceeding installed cabling plant rating
	Length	Per site cable management requirements

Power and Electrical Infrastructure

Power distribution and electrical infrastructure components serving VA telecommunications and data center environments. All electrical work shall comply with the National Electrical Code (NEC), applicable VA standards, and local authority having jurisdiction (AHJ) requirements.

Product Category	Salient Characteristic	Requirement
Rack-Level Power Distribution Units (PDUs)	Type	Metered, switched, or monitored per site requirements
	Voltage/Phase	120V/208V single or three-phase per rack power design
	Monitoring	Network-manageable (SNMP) with per-outlet monitoring preferred
	Bundled Services	Installation, power circuit connection, configuration, labeling
Room-Level Power Distribution	Type	Floor-mount or overhead PDU/RPP per site design

Product Category	Salient Characteristic	Requirement
	Capacity	Sized to site electrical design and load requirements
	Bundled Services	Installation, electrical connection, circuit documentation
Busway Systems and Power Whips	Rating	Per site electrical design
	Bundled Services	Installation, connection, testing, labeling
Grounding and Bonding Infrastructure	Standard	TIA-607-D compliant telecommunications bonding and grounding
	Bundled Services	Design, installation, testing, certification, documentation

Environmental and Cooling Infrastructure

Cooling and environmental monitoring systems supporting VA data center and telecommunications environments.

Product Category	Salient Characteristic	Requirement
Cooling Systems	Type	In-row, overhead, or perimeter CRAC/CRAH units per site design
	Capacity	Sized to site heat load and redundancy requirements

Product Category	Salient Characteristic	Requirement
	Refrigerant	Environmentally compliant per current EPA regulations
	Bundled Services	Site survey, removal of existing unit (if applicable), installation, commissioning, startup, documentation
Hot Aisle / Cold Aisle Containment	Type	Containment curtains, panels, or enclosures per site layout
	Bundled Services	Design, installation, airflow optimization documentation
Environmental Monitoring Sensors	Parameters	Temperature, humidity, airflow, water/leak detection
	Integration	Compatible with VA environmental monitoring platforms
	Bundled Services	Installation, configuration, integration, alarm threshold documentation

Facilities and Life Safety Infrastructure

Fire protection, physical security, and supporting facilities infrastructure for VA telecommunications and data center spaces.

Product Category	Salient Characteristic	Requirement
Fire Detection and Suppression Systems	Detection	Very Early Smoke Detection Apparatus (VESDA) or equivalent
	Suppression	Clean agent (FM-200, Novec 1230, or equivalent) per site design
	Compliance	NFPA 75, NFPA 76, and local AHJ requirements
	Bundled Services	Design, installation, testing, commissioning, AHJ coordination, documentation
Conduit, Sleeves, and Penetrations	Type	EMT, rigid, or flexible per NEC and site requirements
	Firestopping	UL-listed firestop systems for all penetrations
	Bundled Services	Installation, firestopping, pathway documentation
Physical Security Infrastructure	Type	Access control, CCTV, or physical

Product Category	Salient Characteristic	Requirement
		barriers per site requirements
	Bundled Services	Installation, integration with VA security systems, documentation